Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors: Progress Report for Weeks 1–4

Group 8, ECE501

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***Abstract*—**This report presents the first four weeks of a classical computer-vision study to detect and localize defects in fabrics in the AITEX Fabric Image Database (AFID) based on deviation from normal texture. Dataset characterization: The dataset consisted of 141 defect-free and 106 defective 4096×256 images, 107 masks, 12 defect types and 7 fabric codes (used for fabric type) in Week 1. In week 2, image-mask pairs were validated and a set of 3,222 patches were produced from defective images (367 patches with defects) and 4,371 patches were extracted from the defect-free images after CLAHE and Gaussian denoising 256×256 pixels. To create a reference, Weeks built a per-fabric normal-texture, and compared four descriptors: multi-radius uniform LBP, GLCM Haralick features, a Gabor filter bank, and a Gabor+GLCM combination, using the distance to the reference and Otsu thresholding. On a common 10-patch validation sample, LBP best (mean IoU 0.104, F1 0.136), and the other three descriptors (IoU 0.043–0.064, high recall, but low precision) were best on this small sample of images, but are not suitable for use in larger data sets. There was only slight overlap between two of the 10 patches and the ranking is provisional. The LBP F1 score increased from 0.002 to 0.136 when the reference was grouped according to fabric code, rather than image number. The adaptive multi-scale work in Week 5 is motivated by small defects, a fixed window size and a fixed threshold limit accuracy.

***Keywords*—**fabric defect detection; texture descriptors; local binary patterns; gray-level co-occurrence matrix; Gabor filters; AITEX dataset; image segmentation.

I. INTRODUCTION

Manual inspection is time-consuming and unreliable, and the AITEX Fabric Image Database (AFID) [1] supplies pixel level masks for automatic approaches. In this project, instead of modelling the texture, the classical texture descriptors are used: a statistical model is built based on a set of normal images that are extracted from the image using the defect-free method; and the regions that do not match the normal image are determined to be defects. The ten-week plan ends with a comparison of fixed and adaptive multi-scale parameters. This report includes the analysis of the dataset, the pre-processing and patch generation of the dataset, and a fixed-scale comparison of LBP, GLCM and Gabor descriptors in this report, the work is done during Weeks 1-4.

II. METHODOLOGY

A. Dataset analysis (Week 1)

The folders of AFID images (NODefect_images, Defect_images, Mask_images) were checked. Filenames are structured as nnnn_ddd_ff (where nnnn refers to a number, ddd to a defect code and ff to a fabric code). Image sizes were calculated, as well as intensity statistics, defect-type and fabric-wise counts and mask defect-area percentages.Each defective image was compared with its corresponding mask and image sizes, intensity statistics, defect-type and fabric-wise counts, and mask defect-area percentages were computed..

B. Preprocessing and patches (Week 2)

There were 104 images; one was not of good quality and it did not have a mask; the other had a blank mask, so it was not included. Two images were loaded with two disjoint masks, they have been combined together using the logical OR operation. Images were received and processed with CLAHE on the luminance channel and a
light Gaussian blur, and tiled into patches of 256×256 with stride 128; defect free images were tiled similarly.

C. LBP baseline (Week 3)

The uniform LBP histograms at 1, 2 and 3 radiuses were merged together by a 32×32 window. A reference histogram was taken by averaging non-overlapping windows of the defect-free patches, discarding 27,317 near-flat windows (std < 5). Test patches were scored using a sliding window (stride 16) based on chi-square distance to get the deviation map and subsequently Otsu's method was used to get the predicted mask. The score of confidence (separation of deviation inside versus outside the mask) and the IoU, precision, and recall were calculated per pixel as well as F1 score. The reference statistics were constructed separately for each fabric code (the third token in the file name, codes 00–06) based on the patches of that code that are defect-free (i.e., 620–651 of each code)..

D. GLCM and Gabor (Week 4)

GLCM features (contrast, energy, homogeneity, correlation; 8 gray levels, 4 angles) and Gabor energies (4 orientations × 3 frequencies) were computed at the same window scale and combined into a single vector Gabor energies + GLCM. Since these are continuous features, deviation was calculated as a z score Euclidean distance from the mean and standard deviation of the reference sample. We use the same 10 randomly sampled defects containing patches (seed 42) and the same threshold to score all four descriptors.

III. RESULTS

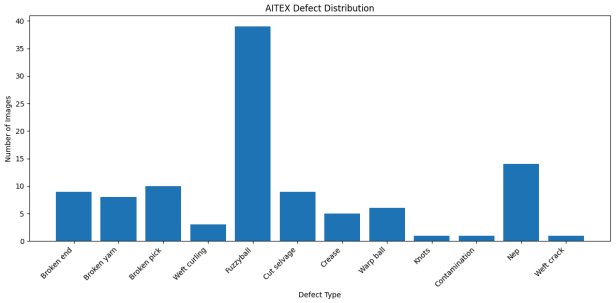
A. Dataset and patches

Table I summarizes the data. There is imbalance among defect types (Fig. 1): fuzzyball contains 39 images, nep 14, while knots, contamination and weft crack contain only one image each. By fabric code, 69 of 106 defective images belong to fabrics 02 (38) and 03 (31). Defects are small: the median mask covers 0.0087% of an image (about 91 of 1,048,576 pixels, less than one 32×32 window), and only 367 of 3,222 patches (11.4%) contain any defect pixel. Mean defect area also differs by fabric, from 0.011% (fabric 06) to 1.92% (fabric 01).

TABLE I
Dataset and Patch Statistics

Item	Value
Defect-free images (7 fabric codes, 00–06)	141
Defective images / mask files	106 / 107
Defect types	12
Image size (pixels)	4096×256 (one 3796×256)
Usable images after exclusions	104
Patches from defective images	3,222 (367 with defects)
Defect-free patches	4,371 (620–651 per fabric)
Defect area: median / mean / max	0.0087% / 0.44% / 22.3%
Mean defect area by fabric (range)	0.011% – 1.92%

Fig. 1. Number of defective images per defect type in AFID.



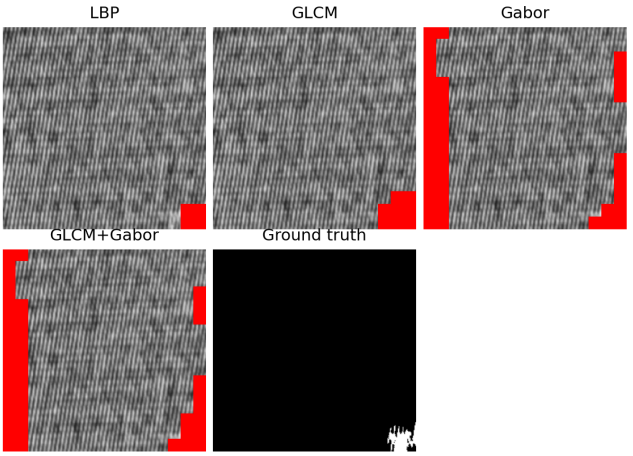
B. Descriptor comparison

Table II shows the descriptors on the 10-patch sample, while Table III shows IoU per patch. LBP ranked first (IoU 0.104, F1 0.136), followed by GLCM+Gabor (0.064, 0.086), Gabor (0.064, 0.085) and GLCM (0.043, 0.068). The three other descriptors had a recall of 0.59 - 0.69 but could only achieve a precision of 0.044 - 0.066, meaning that they can catch a lot of area, but very few defects, while LBP was more selective (precision 0.111, recall 0.175). The average is based on two patches: LBP on 0.010 (broken yarn) achieved IoU 0.616 (Fig. 2), and Gabor and GLCM+Gabor 0.570 and 0.427 on 0.103 (broken pick). On the other eight patches (five fuzzyball, two cut selvage, one nep) LBP scored zero and no descriptor exceeded IoU 0.003; Fig. 3 shows one such miss. The second-highest score (11.8) was for a patch with IoU 0 having no relation to accuracy (Spearman ρ = 0.01 over 40 descriptor–patch results).

TABLE II
Descriptor Comparison (Fixed Scale, 10 Patches)

Descriptor	IoU	Precision	Recall	F1
LBP	**0.104**	**0.111**	**0.175**	**0.136**
GLCM	0.043	0.044	0.656	0.068
Gabor	0.064	0.065	0.592	0.085
GLCM+Gabor	0.064	0.066	0.692	0.086

Fig. 2.



Predicted defect regions (red) of each descriptor on patch 0010_006_02_x2944_y0 (broken yarn), with the ground truth at the bottom right. Gabor-based masks also mark the left and right borders.

TABLE III
IoU per Validation Patch (image number and defect type)

Patch (defect)	LBP	GLCM	Gabor	GLCM+Gabor
0010 broken yarn	0.616	0.343	0.063	0.063
0023 fuzzyball	0.000	0.001	0.003	0.003
0025 fuzzyball	0.000	0.000	0.000	0.001
0032 fuzzyball	0.000	0.000	0.000	0.000
0053 fuzzyball	0.000	0.001	0.001	0.001
0058 fuzzyball	0.000	0.000	0.000	0.000
0064 cut selvage	0.000	0.000	0.000	0.000
0093 nep	0.000	0.003	0.000	0.003
0103 broken pick	0.427	0.086	0.570	0.570
0104 cut selvage	0.000	0.000	0.000	0.000

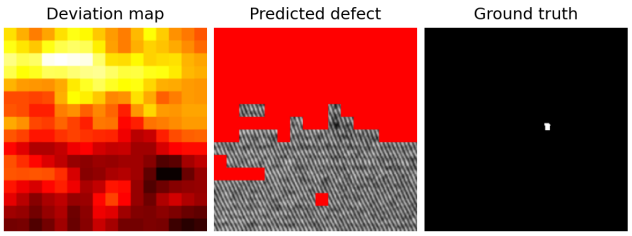


Fig. 3. Failure case for LBP on patch 0023_019_02_x1920_y0 (fuzzyball): the deviation map has no peak at the tiny defect, and Otsu's split flags the upper half of the patch.

IV. DISCUSSIONS

Fabric-specific references. The previous run grouped the reference by image number rather than fabric code, thus combining all seven fabrics into a single group. On the same 10 patches the pooled setting gave LBP an IoU of 0.001 (F1 0.002). With references built per fabric code, LBP rose to 0.104 (F1 0.136), GLCM from 0.005 to 0.043 and Gabor from 0.042 to 0.064, while GLCM+Gabor stayed at 0.065 versus 0.064. This implies that a reference with various weaves will display a different picture of normal texture, and LBP

histograms will be particularly affected by it.

The size of the defect and the window scale. Because the window level score is applied to a fixed window of 1024 pixels and the median defect is about 91 pixels, most defects cannot be isolated by just looking at the window-level score. In Fig. 3 the deviation map gradually changes within the patch and does not have a peak at the defect, so Otsu's method always makes a split in a large area unrelated to the defect. Means the defect area varies by over 100x from fabric to fabric, so a single window size is likely not to be best for everyone, which is a reason to consider adaptive scales.

Gabor border effect. Gabor-based masks are used to delineate the borders of the left and right patches in Fig. 2, which may be due to the border handling effect in the filters and has yet to be confirmed. This reduces their accuracy and could account for their lag behind LBP on that patch.

Limits of the evidence. The 10 defective patches and two patches that provide all the average patches raise only a "tentative" ranking. Image level detection on defect-free patches was not investigated and fabric-wise and defect-wise breakdowns require a larger sample. The previous notebook reviews also addressed missing CLAHE, missing denoising, empty defect-free image list due to nested folders, mismatch in size between reference and scoring windows, a bug in the parsing of the mask-filename for multi-mask images, and alpha channel values in the Week 1 intensity statistics.

V. CONCLUSION

After 4 weeks, a validated dataset was obtained, a patch pipeline was established, and a fixed-scale comparison was created where LBP is the provisional baseline (F1 0.136) but absolute accuracy is low and based on 10 patches. In week 5, all the 367 defect patches will be tested, as well as defect-free patches, and replace Otsu with percentile or adaptive thresholds, and introduce multi-scale windows – the parameters will be selected per fabric, so that fixed and adaptive threshold settings can be compared by type of fabric and type of defect.

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